BEVS Numericals
Formula Sheet

Typeset in LaTeX — every formula in proper mathematical notation

Study-ready formulas for the numericals of CY202: Biology and Environmental Science, taken from *Numericals – 1: Revised Solutions*. Subscripts and superscripts that the source PDF's text layer had corrupted (for example `COI` for CO_2, `SOI` for SO_2) are restored here, and every formula is set as display mathematics so that the symbols, subscripts, fractions and units are unambiguous at print size.

Sections	1–10
Formulas boxed	23
Worked checks	6
Notation	SI, with the source's conventions kept

Notation used throughout

Symbol	Meaning
C	concentration of a pollutant or solute
MW, M	molecular weight ($\mathrm{g\,mol^{-1}}$) / molar mass
DF	dilution factor (dimensionless)
DO_i, DO_f	initial and final dissolved oxygen ($\mathrm{mg\,L^{-1}}$)
EW	equivalent weight ($\mathrm{g\,eq^{-1}}$ or $\mathrm{mg\,meq^{-1}}$)
n-factor	equivalents per mole in the reaction concerned
η	efficiency, as a fraction (35% → 0.35)
LF	load factor of a plant, as a fraction
TH, CH, NCH	total, carbonate and non-carbonate hardness

Conventions kept from the source: molar volume ≈ 22.4 $\mathrm{L\,mol^{-1}}$ at 0 °C and 1 atm; 8760 h in a year; hardness and alkalinity always reported as $\mathrm{mg\,L^{-1}}$ as $\mathrm{CaCO_3}$.

Contents

Notation used throughout 1

1 Gas volume and air-pollutant concentrations 2
1.1 Ideal-gas molar volume . . . 2
1.2 Gas concentration to ppm by volume at STP . . . 2
1.3 Annual vehicle emissions . . . 2

2 Thermal power plant and fuel combustion 2
2.1 Annual generation and fuel use . . . 3
2.2 $\mathrm{SO_2}$, $\mathrm{CO_2}$ and particulate emissions . . . 3

3 Concentration units and chemical calculations 3

4 Gas mixtures and combustion 4

5 Water hardness and equivalent concentrations 4

6 BOD, COD, TOC and theoretical oxygen demand 5

7 pH, carbonate equilibria and alkalinity 5

8 Ion balance and sodium 6

9 Quick constants, conversions and checks 7

10 Source limitations 7

1 Gas volume and air-pollutant concentrations

1.1 Ideal-gas molar volume

Molar volume at a new temperature

$$V_2 = V_1 \times \frac{T_2}{T_1}$$

where V_1, V_2 are the initial and final gas volumes; T_1, T_2 the absolute temperatures.
Use when: pressure and the amount of gas are constant. Temperatures must be in kelvin.

Celsius to kelvin

$$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273$$

Checks. At 25 °C:

$$22.4 \times \frac{298}{273} \approx 24.5\ \mathrm{L\,mol^{-1}}, \qquad \text{at } 100\,^\circ\mathrm{C}: \quad 22.4 \times \frac{373}{273} \approx 30.6\ \mathrm{L\,mol^{-1}}$$

Note: the handwritten 100 °C value appears as 104.5 $\mathrm{L\,mol^{-1}}$ in the scan; the revised PDF flags it as inconsistent. At 1 atm the relationship above gives ≈ 30.6 $\mathrm{L\,mol^{-1}}$.

1.2 Gas concentration to ppm by volume at STP

mg m^{-3} → ppmv

$$\mathrm{ppmv} = \frac{C\,(\mathrm{mg\,m^{-3}}) \times 22.4}{\mathrm{MW}\,(\mathrm{g\,mol^{-1}})}$$

where 22.4 $\mathrm{L\,mol^{-1}}$ is the molar volume at STP.
Use when: converting a mass concentration of a gas to a volume fraction; convert $\mu\mathrm{g\,m^{-3}}$ to $\mathrm{mg\,m^{-3}}$ first, since 1000 μg = 1 mg.

Checks.

Gas	C	MW	ppmv
CO	30 $\mathrm{mg\,m^{-3}}$	28	$30 \times 22.4/28 = 24.0$
NO_2	400 $\mu\mathrm{g\,m^{-3}} = 0.400\ \mathrm{mg\,m^{-3}}$	46	$0.400 \times 22.4/46 \approx 0.195$
SO_2	350 $\mu\mathrm{g\,m^{-3}} = 0.350\ \mathrm{mg\,m^{-3}}$	64	$0.350 \times 22.4/64 \approx 0.123$

1.3 Annual vehicle emissions

Fleet distance, mass and tonnes

$$\text{annual distance} = (\text{number of vehicles}) \times (\text{distance per vehicle per year})$$

$$\text{annual mass} = (\text{annual distance}) \times (\text{emission factor})$$

$$\text{tonnes} = \frac{\text{mass in grams}}{10^6}$$

where an emission factor in $\mathrm{g\,km^{-1}}$ gives grams when multiplied by kilometres.

For solvent or paint problems the same logic applies to use rather than distance:

$$\text{total annual use} = (\text{number of users or vehicles}) \times (\text{annual use per unit})$$

2 Thermal power plant and fuel combustion

2.1 Annual generation and fuel use

Electricity output

$$E\,(\mathrm{MWh\,yr^{-1}}) \;=\; P\,(\mathrm{MW}) \times 8760\ \mathrm{h\,yr^{-1}} \times \mathrm{LF}$$

Use when: 8760 h assumes a 365-day year; convert MWh to kWh by multiplying by 10^3.

Fuel requirement

$$\text{fuel}\ (\mathrm{kg\,yr^{-1}}) = \frac{\text{annual energy output}\ (\mathrm{kWh\,yr^{-1}})}{\text{net energy yield}\ (\mathrm{kWh\,kg^{-1}})}$$

$$\text{fuel energy input} = \frac{\text{electrical energy output}}{\eta}$$

$$\text{fuel mass} = \frac{\text{fuel energy input}}{\text{calorific value}}$$

where η is the thermal efficiency as a fraction and the calorific value is in energy per unit mass.

Source example. For a 915 MW plant at a load factor of 0.70:

$$915 \times 8760 \times 0.70 = 5{,}606{,}820\ \mathrm{MWh\,yr^{-1}}$$

2.2 SO_2, CO_2 and particulate emissions

Stoichiometric emission masses

$$m_{SO_2} = m_S \times \frac{64}{32} = 2\,m_S \qquad\qquad m_{CO_2} = m_C \times \frac{44}{12} \approx 3.667\,m_C$$

where 64 and 32 are the molar masses of SO_2 and S; 44 and 12 those of CO_2 and C.
Use when: combustion is complete and no retention or removal is specified.

Carbon in the fuel, and particulate control

$$m_C = (\text{fuel mass}) \times \frac{\text{carbon }\%}{100}$$

$$\text{uncontrolled particulate} = (\text{fuel throughput}) \times (\text{particulate generation factor})$$

$$\text{controlled emissions} = (\text{uncontrolled}) \times (1 - \eta_c)$$

where η_c is the collection efficiency as a fraction — 99.5% means $\eta_c = 0.995$, so the residual fraction is 0.005.

3 Concentration units and chemical calculations

Mass fraction, ppm by mass and molarity

$$\text{weight }\% = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100$$

$$\text{ppm by mass} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$$

$$M\,(\mathrm{mol\,L^{-1}}) = \frac{\text{moles of solute}}{\text{volume of solution (L)}}$$

$$n = \frac{m\,(\mathrm{g})}{M\,(\mathrm{g\,mol^{-1}})}$$

where the denominator of a mass fraction is the *total solution* mass, not the solvent mass.
Use when: for dilute aqueous solutions only, 1 ppm $\approx$ 1 $\mathrm{mg\,L^{-1}}$ (density $\approx$ 1 $\mathrm{kg\,L^{-1}}$).

Mole fraction, equivalents and normality

$$x_i = \frac{n_i}{\sum_j n_j}, \qquad \sum_i x_i = 1$$

$$\text{equivalents} = \frac{m\,(\text{g})}{\text{EW}\,(\text{g eq}^{-1})}$$

$$N\,(\text{eq L}^{-1}) = \frac{\text{equivalents}}{\text{volume of solution (L)}}$$

$$\text{EW} = \frac{M}{n\text{-factor}}$$

where the n-factor is the number of H^+, OH^- or charge equivalents per mole in the reaction concerned.
Use when: the equivalent weight must match the reaction — it is not a fixed property of a substance.

Stoichiometry workflow. Balance the equation → convert the known mass to moles → apply the mole ratio → convert the required moles to mass, volume or concentration → check units and significant figures.

4 Gas mixtures and combustion

Average molecular weight of a mixture

$$\overline{M} = \sum_i y_i M_i$$

where y_i is the mole (or volume) fraction of component i and M_i its molecular weight.
Use when: volume percentages must first be converted to fractions, and the fractions must sum to 1. At the same temperature and pressure, volume fraction ≈ mole fraction.

Combustion mass ratios

$C + O_2 \rightarrow CO_2$	12 kg C → 44 kg CO_2
$S + O_2 \rightarrow SO_2$	32 kg S → 64 kg SO_2
$2\,H_2 + O_2 \rightarrow 2\,H_2O$	4 kg H_2 → 36 kg H_2O

Note: for incomplete combustion or excess-air problems, use the stated fuel composition, air requirement, excess-air percentage and product analysis. Do not assume complete conversion unless it is stated.

5 Water hardness and equivalent concentrations

Equivalent weight of an ion

$$\text{EW} = \frac{\text{atomic or molecular weight}}{\text{ionic charge}}$$

Conversion to $CaCO_3$ equivalents

$$C\,(\text{mg L}^{-1}\ \text{as CaCO}_3) = C_{\text{ion}}\,(\text{mg L}^{-1}) \times \frac{50}{\text{EW}}$$

$$\text{meq L}^{-1} = \frac{C\,(\text{mg L}^{-1})}{\text{EW}\,(\text{mg meq}^{-1})}$$

$$\text{hardness as CaCO}_3 = \text{meq L}^{-1} \times 50$$

where 50 is the equivalent weight of $CaCO_3$ used throughout the source.

Calcium, magnesium and total hardness

$$\text{Ca hardness} = [\text{Ca}^{2+}] \times \frac{50}{20.04}$$

$$\text{Mg hardness} = [\text{Mg}^{2+}] \times \frac{50}{12.15}$$

$$\text{TH} = \text{Ca hardness} + \text{Mg hardness}$$

$$\text{CH} = \min(\text{TH},\ \text{total alkalinity}), \qquad \text{NCH} = \text{TH} - \text{CH}$$

where all hardness and alkalinity values are on the same $CaCO_3$-equivalent basis.

Ion	Equivalent weight	Factor 50/EW	Example
Ca^{2+}	$40.08/2 = 20.04$	2.50	60 mg L^{-1} → 150 mg L^{-1} as $CaCO_3$
Mg^{2+}	$24.3/2 = 12.15$	4.12	29.3 mg L^{-1} → 121 mg L^{-1} as $CaCO_3$
HCO_3^-	≈ 61	0.82	366 mg L^{-1} → 300 mg L^{-1} alkalinity

6 BOD, COD, TOC and theoretical oxygen demand

Biochemical oxygen demand

$$\text{BOD}\,(\text{mg L}^{-1}) \;=\; (\text{DO}_i - \text{DO}_f) \times \text{DF}$$

where DO_i and DO_f are the initial and final dissolved oxygen concentrations.
Use when: the source tests BOD over 5 days at 20 °C; apply the seed/blank correction if one is specified.

Theoretical oxygen demand — ethylene glycol

$$\text{C}_2\text{H}_6\text{O}_2 + 2.5\,\text{O}_2 \rightarrow 2\,\text{CO}_2 + 3\,\text{H}_2\text{O}$$

$$\text{ThOD} \;=\; \frac{2.5 \times 32}{62}\,C \;\approx\; 1.29\,C \;\; \text{mg L}^{-1}$$

where C is the ethylene glycol concentration in mg L^{-1} and 62 g mol^{-1} its molar mass.

Theoretical oxygen demand — phenol

$$\text{C}_6\text{H}_6\text{O} + 7\,\text{O}_2 \rightarrow 6\,\text{CO}_2 + 3\,\text{H}_2\text{O}$$

$$\text{ThOD} \;=\; \frac{7 \times 32}{94}\,C \;\approx\; 2.38\,C \;\; \text{mg L}^{-1}$$

where C is the phenol concentration in mg L^{-1} and 94 g mol^{-1} its molar mass.

Checks. 100 mg L^{-1} ethylene glycol → ≈ 129 mg L^{-1} O_2; 120 mg L^{-1} phenol → ≈ 286 mg L^{-1} O_2.

BOD, COD and TOC measure different things. COD is measured chemically with potassium dichromate (about 2 h digestion) and is generally ≥ BOD; TOC measures the carbon in organic compounds, with inorganic carbon removed or accounted for separately. There is no universal conversion factor between the three — the relationship depends on the sample and its biodegradability.

7 pH, carbonate equilibria and alkalinity

pH

$$\text{pH} \;=\; -\log_{10}\left[\text{H}^+\right]$$

Carbonate system

$$\mathrm{CO_2(aq) + H_2O \;\rightleftharpoons\; H_2CO_3^{*} \;\rightleftharpoons\; H^+ + HCO_3^-}$$

$$K_1 = \frac{[\mathrm{H^+}][\mathrm{HCO_3^-}]}{[\mathrm{H_2CO_3^{*}}]}$$

$$K_2 = \frac{[\mathrm{H^+}][\mathrm{CO_3^{2-}}]}{[\mathrm{HCO_3^-}]}$$

where $\mathrm{H_2CO_3^{*}}$ is the dissolved $\mathrm{CO_2}$ plus true carbonic acid, as in the source.

Alkalinity

$$\text{alkalinity}\,(\mathrm{mg\,L^{-1}\ as\ CaCO_3}) \;=\; \sum_i C_i \times \frac{50}{\mathrm{EW}_i}$$

where the principal contributors in the source are $\mathrm{HCO_3^-}$, $\mathrm{CO_3^{2-}}$ and $\mathrm{OH^-}$.

Species	Term	Example
$\mathrm{HCO_3^-}$	$C \times 50/61$	$260\ \mathrm{mg\,L^{-1}} \rightarrow 213\ \mathrm{mg\,L^{-1}}$ as $\mathrm{CaCO_3}$
$\mathrm{CO_3^{2-}}$	$C \times 50/30$	$30\ \mathrm{mg\,L^{-1}} \rightarrow 50\ \mathrm{mg\,L^{-1}}$ as $\mathrm{CaCO_3}$
$\mathrm{OH^-}$	$C \times 50/17$	—

Note: pH alone does not fix the bicarbonate concentration — temperature, the equilibrium constants, total inorganic carbon and the species assumptions are all needed. In the source's rainwater example (pH ≈ 5.65), the $\mathrm{OH^-}$ and $\mathrm{CO_3^{2-}}$ contributions are treated as negligible.

8 Ion balance and sodium

Charge balance

$$\sum \text{cations}\,(\mathrm{meq\,L^{-1}}) \;\approx\; \sum \text{anions}\,(\mathrm{meq\,L^{-1}})$$

where every ion must be converted to $\mathrm{meq\,L^{-1}}$ before summing.
Use when: checking an analysis; a large mismatch means an ion has been omitted or a datum is wrong.

Sodium from the balance

$$[\mathrm{Na^+}]\,(\mathrm{meq\,L^{-1}}) = \sum \text{anions} - \sum \text{known cations}$$

$$[\mathrm{Na^+}]\,(\mathrm{mg\,L^{-1}}) = [\mathrm{Na^+}]\,(\mathrm{meq\,L^{-1}}) \times 23$$

where 23 $\mathrm{mg\,meq^{-1}}$ is the sodium factor used in the source.

9 Quick constants, conversions and checks

Item	Value	Use / condition
Molar volume, 0 °C, 1 atm	22.4 $\mathrm{L\,mol^{-1}}$	STP convention of the source
Kelvin conversion	$T(\mathrm{K}) \approx T(^\circ\mathrm{C}) + 273$	all gas-volume work
Molar volume, 25 °C	24.5 $\mathrm{L\,mol^{-1}}$	from $22.4 \times 298/273$
Molar volume, 100 °C	30.6 $\mathrm{L\,mol^{-1}}$	at 1 atm, from $22.4 \times 373/273$
Mass conversions	1000 μg = 1 mg; 1000 mg = 1 g; 1000 g = 1 kg	convert before substituting
Metric tonne	1 t = 1000 kg = 10^6 g	annual emissions
Hours per year	8760 h	assumes 365 days
Combustion ratios	$CO_2/C = 44/12$; $SO_2/S = 64/32$; $H_2O/H_2 = 36/4$	balanced reactions
Ca^{2+} equivalent weight	$40.08/2 \approx 20.04$	hardness
Mg^{2+} equivalent weight	$24.3/2 \approx 12.15$	hardness
$CaCO_3$ factor	50/EW	$\mathrm{mg\,L^{-1}}$ as $CaCO_3$
Sodium factor	23 $\mathrm{mg\,meq^{-1}}$	Na^+ $\mathrm{meq\,L^{-1}}$ to $\mathrm{mg\,L^{-1}}$
BOD test	5 days at 20 °C	apply stated corrections
COD test	$K_2Cr_2O_7$, about 2 h digestion	as described in the source

10 Source limitations

- Several handwritten inputs in the scan — vehicle counts, travel distances, emission factors, fuel-analysis values, gas-mixture fractions and the alum-treatment quantities — are faint or ambiguous. They are not filled in by guesswork here.
- The 100 °C molar volume is recalculated as $\approx 30.6\ \mathrm{L\,mol^{-1}}$ at 1 atm, consistent with the ideal-gas temperature relationship; the 104.5 $\mathrm{L\,mol^{-1}}$ in the handwriting is inconsistent.
- The theoretical oxygen-demand examples use the balanced equations and concentrations stated in the revised PDF; substitute the exact problem values if they differ.
- Page references throughout refer to the 11-page *Numericals – 1: Revised Solutions*: overview and quick summary on pp. 1 and 11, topic formulas on pp. 2–10.

How to use this sheet. Learn the boxed relationship first, then the "use when" line, because the mark in a numerical usually goes to the correct substitution and unit rather than to the arithmetic. Every formula here is dimensionally consistent: keep kilometres with km, grams with g, and convert to tonnes or kilograms only at the last step.